

1. Administrative & Study Identification

Full Study Title: Safety and Efficacy of Lorbrena **Short Title/**
Acronym: Lorbrena PMS (Japan) **Protocol Number:** B7461018 **NCT**
Number: NCT03844464 **Sponsor Name:** Pfizer **Investigational Product:**
Lorbrena (lorlatinib) **Phase of Development:** N/A (Observational /
Post-marketing) **Medical Monitor:** Pfizer Post-Marketing
Surveillance Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To investigate the safety and efficacy factors under post-marketing use of Lorbrena in patients who received this drug, specifically factors affecting the onset of central nervous system (CNS) disorders, and the effect of Lorbrena in combination with CYP3A inducers on the onset of hepatic dysfunction. **Secondary Objectives:** To continuously assess the general safety and efficacy profile of Lorbrena in a real-world clinical setting.

2.2 Background & Rationale To fulfill post-marketing requirements and monitor the real-world safety profile of Lorbrena (lorlatinib) in patients with ALK-positive non-small cell lung cancer (NSCLC). This study ensures that specific adverse events—such as CNS side effects and potential drug-drug interactions (e.g., with CYP3A inducers)—are fully characterized and understood outside of controlled clinical trials.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Post-marketing special investigation) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target** **\$\$\$:** 1,290 participants
Number of Sites: Multiple clinical sites across Japan **Estimated**
Study Start: 21-Jan-2019 **Estimated Primary Completion:** 30-Jul-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients diagnosed with Non-small Cell Lung Cancer (NSCLC). 2. All administered patients receiving Lorbrena (lorlatinib) tablets in a real-world clinical setting.

4.2 Exclusion Criteria 1. Patients not receiving Lorbrena. 2. Any specific contraindications according to the local approved label (e.g., patients with a history of severe hypersensitivity to lorlatinib).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: As prescribed by the treating physician per the approved package insert for Lorbrena. **Route of Administration:** Oral **Duration of Treatment:** Real-world duration (determined by clinical benefit, disease progression, or physician decision).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant therapies as per real-world practice for NSCLC comorbidities. **Prohibited/Caution:** Co-administration with CYP3A inducers must be carefully monitored or avoided due to established risks of hepatic dysfunction.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Baseline	Day 0	Patient characteristics, medical history, Lorbrena initiation
Interim		Standard clinical visits
End of Study		Monitoring for CNS effects, hepatic enzymes, tumor response
Study Completion		Final safety, survival outcomes, and post-marketing reporting

(Note: As an observational post-marketing study, visits follow standard clinical practice schedules)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, severity, and factors affecting the onset of central nervous system (CNS) disorders, as well as hepatic dysfunction when used in combination with CYP3A inducers. **Measurement Method:** Post-marketing safety reports and routine clinical laboratory/neurological assessments.

7.2 Secondary & Exploratory Endpoints Real-world Overall Survival (OS) and Progression-Free Survival (PFS). General post-marketing adverse event profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of post-marketing real-world data (RWD) via physician reports. **Serious Adverse Events (SAEs):** Routine post-marketing pharmacovigilance reporting per

local regulatory timelines. **Data Monitoring Committee (DMC):** Internal pharmacovigilance and safety team at Pfizer.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All administered patients with available follow-up data. **Sample Size Justification:** Target of $N=1,290$ to provide adequate power for detecting rare but significant post-marketing safety signals (CNS and hepatic). **Handling of Missing Data:** Standard observational methodologies for real-world data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Post-marketing Study Practice (GPSP) and local Japanese regulations. **Monitoring Plan:** Centralized review of post-marketing report forms and safety signals. **Audit Trail:** Validated electronic data capture for pharmacovigilance tracking.

11. Study Identifiers & Governance

Primary ID: B7461018 **Registry Number:** NCT03844464 **Sponsor/Lead Organization:** Pfizer **Study Title:** Safety and Efficacy of Lorbrena **Official Regulatory Title:** Special Investigation for LORBRENA Tablets

12. Clinical Framework (NSCLC Specific)

Primary Condition: Non-small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** ALK-positive metastatic NSCLC **Medical Methodology:** Targeted ALK Inhibitor Therapy **Optimization Target:** Manageable toxicity profile (CNS and hepatic) with optimal survival benefit

13. Study Procedures & Methodology

Management Pathway: Standard real-world oncology clinical pathway. **Education Protocol (HCP):** Guidelines on monitoring CNS adverse events and CYP3A interactions. **Education Protocol (Patient):** Counseling on potential CNS symptoms (e.g., mood changes, cognitive effects). **Quality Audit Mechanism:** Pharmacovigilance safety signal detection.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until study completion (July 2025) or treatment discontinuation. **Onsite Visit Cadence:** As per standard

of care in real-world practice. **Remote Monitoring Frequency:**
Continuous pharmacovigilance data sweeps. **Checklist Review**
Interval: Periodic Safety Update Reports (PSURs).

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				